EXPERIMENT 3

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PYTHONCODE:

```
import tkinter as tk
fromtkinterimport ttk, messagebox

def convert():
   try:
      value=float(entry_value.get())
      category=combo_category.get()
      conversion=combo_conversion.get()

      if category== "Length":
         ifconversion=="KilometertoMeter":
            result=value *1000
            output_label.config(text=f"Result:{result} meters")

         elifconversion=="MetertoKilometer":
            result=value / 1000
            output_label.config(text=f"Result:{result} kilometers")

      elif category =="Mass":
         if conversion== "Kilogram to Gram":
            result=value *1000
            output_label.config(text=f"Result:{result} grams")

         elifconversion=="GramtoKilogram":
            result=value / 1000
            output_label.config(text=f"Result:{result} kilograms")

      elif category =="Temperature":
         ifconversion=="CelsiustoFahrenheit":
            result=(value* 9/5) + 32
```

```
            output_label.config(text=f"Result: {result} °F")

        elif conversion == "Fahrenheit to Celsius":
            result = (value - 32) * 5/9
            output_label.config(text=f"Result: {result} °C")

    else:
        messagebox.showerror("Error", "Invalid selection!")

except ValueError:
    messagebox.showerror("Error", "Please enter a valid number!")

def update_options(event):
    category = combo_category.get()

    if category == "Length":
        combo_conversion['values'] = (
            "Kilometer to Meter",
            "Meter to Kilometer"
        )

    elif category == "Mass":
        combo_conversion['values'] = (
            "Kilogram to Gram",
            "Gram to Kilogram"
        )

    elif category == "Temperature":
        combo_conversion['values'] = (
            "Celsius to Fahrenheit",
            "Fahrenheit to Celsius"
        )

    combo_conversion.current(0)

# Create main window
root = tk.Tk()
```

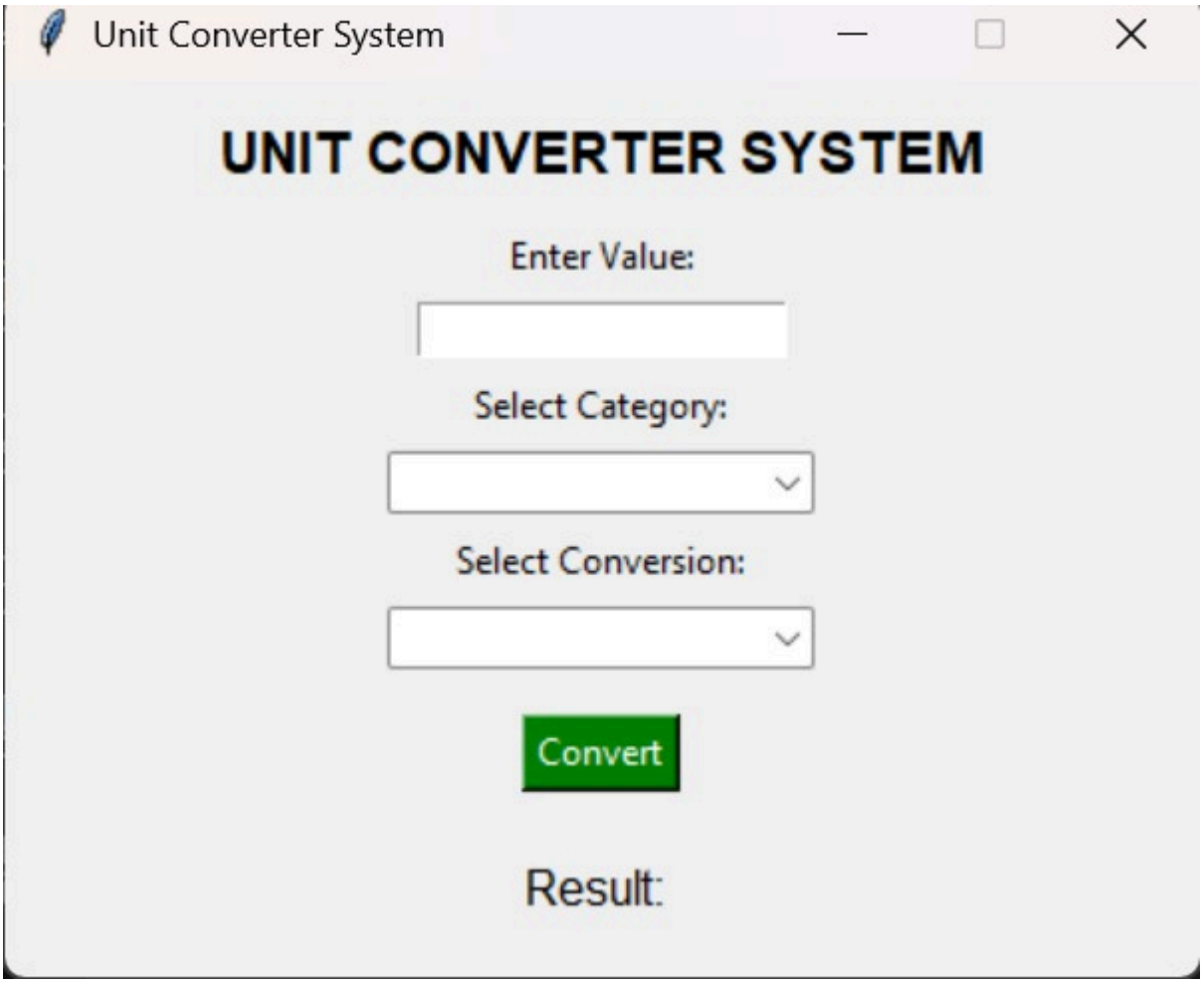
```
root.title("Unit  Converter  System")
root.geometry("400x300")
root.resizable(False, False)

# Title
tk.Label(
    root,
    text="UNIT CONVERTER SYSTEM",
    font=("Arial", 14, "bold")
).pack(pady=10)

# Input value
tk.Label(root, text="Enter Value:").pack()

entry_value = tk.Entry(root, width=20)
entry_value.pack(pady=5)

# Category selection
tk.Label(root, text="Select Category:").pack()

combo_category = ttk.Combobox(
    root,
    state="readonly",
    values=("Length", "Mass", "Temperature")
)
combo_category.pack(pady=5)

combo_category.bind("<<ComboboxSelected>>", update_options)

# Conversion selection
tk.Label(root, text="Select Conversion:").pack()

combo_conversion = ttk.Combobox(root, state="readonly")
combo_conversion.pack(pady=5)

# Convert button
tk.Button(
```

```
    root,
    text="Convert",
    command=convert,
    bg="green",
    fg="white"
).pack(pady=10)

#Output label
output_label = tk.Label(
    root,
    text="Result: ",
    font=("Arial", 12)
)
output_label.pack(pady=10)

#Run program
root.mainloop()
```

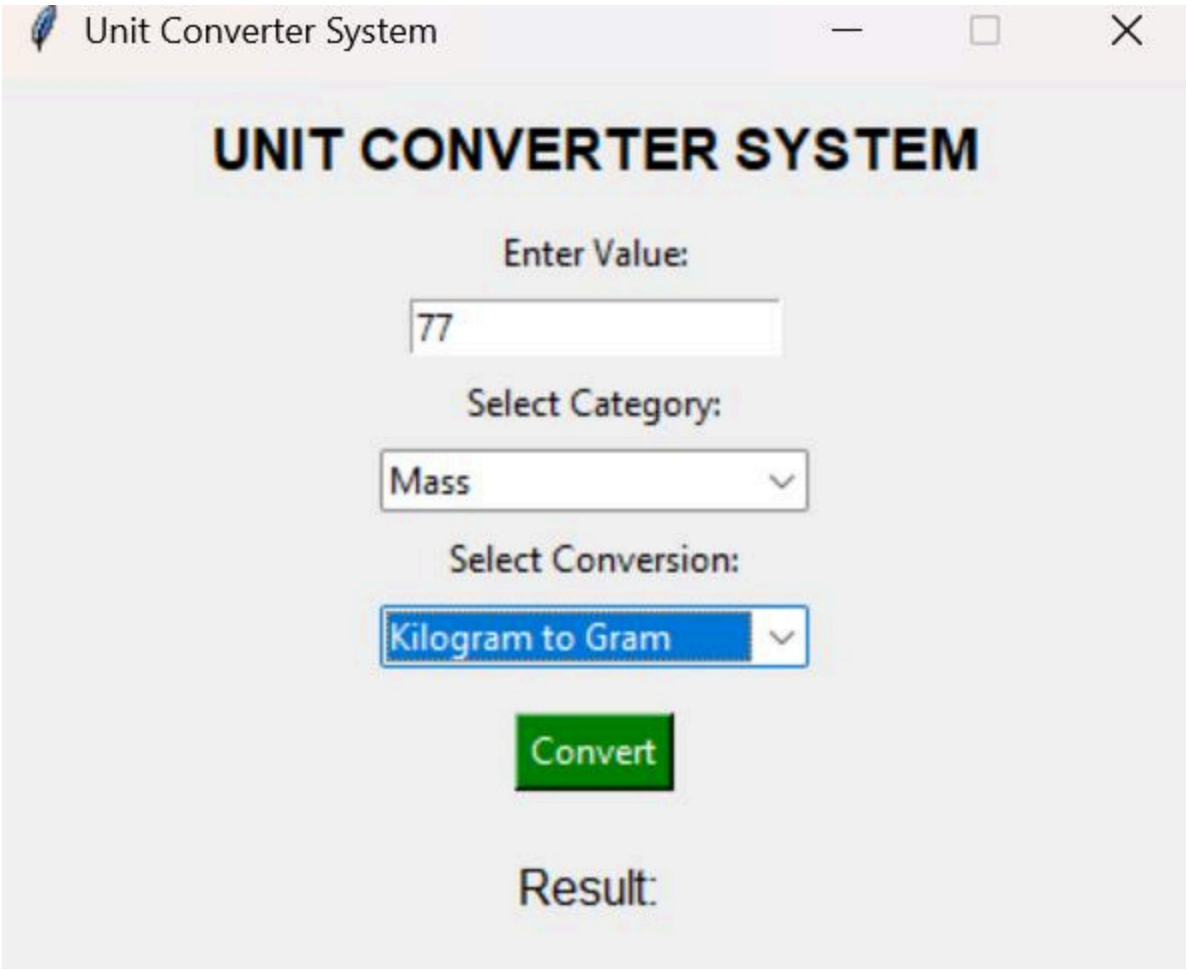
convert() Function — Main Logic

def convert():

This function runs when the Convert button is clicked.



Step 1: Read User Input
value = float(entry_value.get())
    Gets number entered by user
    Converts text → float number
Ifthe user enters invalid data → error handled using:
except ValueError:



Step 2: Get Selected Options

```
category = combo_category.get()
conversion = combo_conversion.get()
```

Example:

Category → Length

Conversion → Kilometer to Meter

Unit Converter System — Explanation

This program creates a GUI (Graphical User Interface) using Python's Tkinter library.

It allows the user to:

Enter a number
Select a unit category
Choose a conversion type
Click Convert
View the converted result

Program Structure

The program has 3 main parts:

1. Import libraries
2. Functions (logic)
3. GUI design

Import Libraries

```
import tkinter as tk
from tkinter import ttk, messagebox
```

tkinter

Used to create windows, buttons, labels, etc.

ttk

Provides styled widgets like Combobox.

messagebox

Displays popup error messages.

GUI Creation (Window Design)

Main Window

```
root = tk.Tk()
```

Creates application window.

```
root.title("Unit Converter System")
root.geometry("400x300")
```

Sets title and size.

Title Label
tk.Label(root,text="UNIT CONVERTER SYSTEM")
Displays heading.

Input Field
entry_value = tk.Entry(root)
User types value here.

Category Combobox
combo_category = ttk.Combobox(...)
Options:
- Length
- Mass
- Temperature

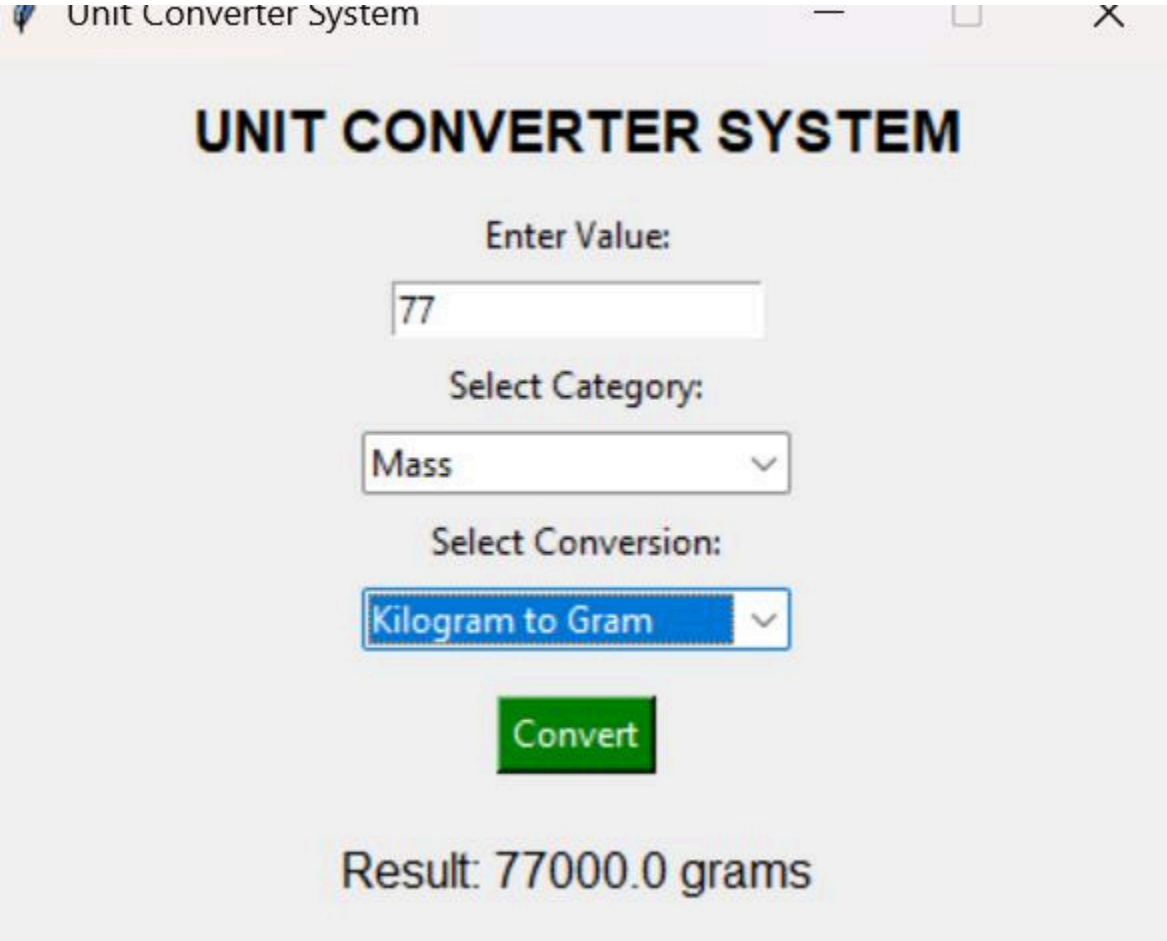
Event binding:
combo_category.bind("<<ComboboxSelected>>", update_options)
Automatically updates conversion choices.

Conversion Combobox
combo_conversion = ttk.Combobox(...)
Shows conversions based on category.

Convert Button
tk.Button(command=convert)
When clicked:
Calls convert() function.

Output Label
output_label = tk.Label(...)
Displays result.

Start Application
root.mainloop()
Runs the GUI event loop continuously.



Program Working Flow

Start Program

↓

Window Opens

↓

User enters value

↓

Select Category

↓

Conversion options update

↓

Click Convert

↓

Calculation happens

↓

Result displayed

Example Execution

Input:

Value = 5

Category = Length

Conversion = Kilometer to Meter

Output:

Result: 5000 meters

Features of Your Program

- User-friendly GUI
- Dynamic dropdown options
- Error handling
- Multiple unit categories
- Real-time result display

Possible Improvements (Optional)

You can enhance by adding:

- Time conversion
- Area conversion
- Currency converter
- Clear button
- Dark theme
- Auto result update